

08:26 pm

**PDF AI Assistant**

ChromaDB RAG + TFLite Engine

● FastAPI Backend Live

☰ Key Takeaways

- ✓ Multimodal Learning in Artificial Intelligence (AI) Last Updated :5 Jul, 2024 - - - Multimodal AI refers to artificial intelligence systems that integrate and process multiple types of data, such as text, images, audio, and video, to understand and generate comprehensive insights and responses.
- ✓ - Conclusion - Multimodal AI - FAQs Overview of Multimodal AI Models Multimodal AI can be defined as Artificial Intelligence systems that can process various types of information ranging from text, images, audio videos, etc.
- ✓ Runway Gen-2 Runway Gen-2 is a multimodal generative AI model from Runway that can create images, videos, and 3D content from text prompts.
- ✓ Parameters Multimodal AI Unimodal AI Integration Integrates information from various modalities Works within a single modality Complexity Higher complexity due to multisource processing Simpler, focusing on one data source Contextual Understanding Enhanced understanding through cross-modal context Limited to the context within one modality Application Scope Broad applications in diverse fields Narrower, specific to the modality User Interaction More natural and intuitive (e.g., voice and gesture) Limited to one mode of interaction Accuracy Potentially higher accuracy due to comprehensive data Accuracy depends on single data type Learning Requires more advanced algorithms for multimodal learning Relatively simpler learning algorithms Data Fusion Uses techniques to merge data from different sources No need for data fusion Processing Power Requires greater computational resources Generally less resourceintensive Real-world Application More aligned with how humans perceive and interact Less aligned with natural human interaction
- ✓ Parameters Multimodal AI Unimodal AI Research Focus Interdisciplinary, involving multiple fields Often specialized within a single field Challenges Data alignment, synchronization, and fusion Limited challenges within one data type What technologies are associated with multimodal AI?

